
Response Geometry in Integrated Gradients: A Diffusion-Policy Case Study

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Abstract

Attribution evaluation depends on the ranking that selects interventions and the response used to score them. We study this distinction for Integrated Gradients (IG) of a fixed-noise diffusion policy’s predicted action chunk. For quadratic and stabilized negative- ℓ_2 action-discrepancy targets, we derive an exact relation between the normalized responses and a positive, path-dependent factor between their gradients. A smooth construction shows that the IG rankings can reverse even when the nonzero gradients are pointwise collinear. A finite-grid example illustrates the response dependence of random-ranking AUC. We examine the practical effect by reanalyzing 99 preserved Robotics Diffusion Transformer files. Changing only the response changes median vision deletion AUC from 0.4479 to 0.2908 while the recorded quadratic-target rankings stay the same. Across eight paired rescore cases, every median response change is positive, but overshooting tails make six means negative. Exploratory episode-omission checks leave the insertion-mean signs unchanged, although the language-deletion signs can reverse. A second smooth construction shows that several integration grids can agree in completeness, ranking and perturbation curves while reversing the exact IG order. In this example, the inaccurate order even scores better on both perturbation metrics. Existing actual-model diagnostics also show why repeatability and integration-budget sensitivity must be considered separately, although the cohort is incomplete and the sensitivity remains unresolved. These results separate the roles of response geometry, numerical accuracy and ranking quality. They do not qualify a production numerical setting, demonstrate superior attribution rankings, or explain task success or learned-weight dependence. The artifact preserves the raw-record lineage, analytic checks, adverse results and provenance limits.

1 Introduction

Vision-language-action (VLA) models such as RT-2 (Zitkovich et al., 2023), Octo (Octo Model Team et al., 2024), and the Robotics Diffusion Transformer (RDT) (Liu et al., 2025) use sensory observations and instructions to predict robot actions. Diffusion policies (Chi et al., 2023) produce action chunks through iterative denoising (Ho et al., 2020). The resulting output is a vector, so applying a scalar attribution method requires choosing which aspect of the action prediction to explain. This choice also affects how the explanation is evaluated.

Integrated Gradients (IG) (Sundararajan et al., 2017) integrates the gradients of a scalar output along a path from a baseline to an input. For a fixed-noise action sample, this scalar can measure the discrepancy from a detached reference action. A quadratic discrepancy and a negative norm both measure change, but they define different scalar functions. Evaluating the attribution requires a second choice. The attribution ranking selects which features to remove or insert, and a response scores those interventions. If a comparison changes both the scalar target and the response, its native-response AUC alone cannot identify an improvement in the ranking.

We study this problem using exact derivations, constructive examples and a retrospective RDT case study. Changing the response from a quadratic discrepancy to a stabilized negative ℓ_2 discrepancy changes normalized AUC even when the selected features and perturbed outputs stay fixed. At the same time, the positive

factor relating the targets’ gradients varies along the path, which allows their IG rankings to differ. The first operation rescores existing interventions, while the second recomputes the attributions. Neither result establishes which ranking is more useful under a common response.

A separate question is whether the computed attribution accurately approximates the specified mathematical quantity. Small scalar completeness residuals can hide errors in individual coordinates. Repeating the same finite-precision calculation can also reproduce the same inaccurate approximation. The smooth construction in Appendix B shows that several nested grids can agree exactly in completeness, ranking and perturbation responses while reversing the exact IG ranking. We therefore report the numerical diagnostics already collected together with the full planned roster and its unfinished cells. Successful checks from part of this roster do not certify the implementation. The intended attribution measures action discrepancy at a fixed context. It does not explain eventual task success or establish safe robot behavior.

Prior work already shows that random-order AUC depends on interactions and that an attribution ordering need not optimize a perturbation score (Hama et al., 2023). We examine these concerns for quadratic and stabilized negative-norm action discrepancies. We make three contributions.

1. **Response and path-weight analysis.** We derive an exact normalized-response identity and state its overshoot and zero-gap qualifications. We also construct a finite-stabilizer example in which the quadratic and negative-norm targets reverse IG feature order. A 100-feature equal-contribution policy provides a calibration example for these responses on the historical evaluation grid.
2. **An authenticated retrospective case study.** We rescore preserved RDT perturbation curves while keeping each recorded ranking fixed. The analysis records the occurrence rules, episode grouping and links from results to source records. Unknown historical context identities prevent a paired comparison of ranking quality.
3. **Numerical diagnostic limits.** We give a smooth counterexample to certification by finite-grid agreement and report fixed-input repeatability and integration-budget checks from a partially completed campaign. Failures and incomplete coverage remain visible, and no production setting is approved.

These contributions separate the issues and document their consequences in one diffusion-policy family. A common-response ranking study would answer the additional question of ranking efficacy. Its absence limits that conclusion without invalidating the analysis that holds rankings fixed.

2 Related Work

2.1 Attribution Methods

Sundararajan et al. (2017) introduced IG with the sensitivity and implementation-invariance axioms and showed that straight-line path integration satisfies both. Later studies address limitations of the original method. Jha et al. (2021) computed attributions over neural stochastic differential equation models to produce smoother attribution maps. Their later work showed that shaping the injected noise further improves robustness (Jha et al., 2022). Walker et al. (2024) found that gradients from saturation regions contribute noise and proposed Integrated Decision Gradients, which concentrates integration on the decision-relevant portion of the path. For vision transformers, metric-driven attributions (Walker et al., 2025b) and information-flow decompositions (Walker et al., 2025a) improve patch-level explanations. Guided IG (Kapishnikov et al., 2021) changes the integration path to reduce pixel-level noise.

Other work directly examines sensitivity to the integration budget. Makino et al. (2024) study completeness residuals that vary across instances, nonmonotone changes with step count and unresolved path gradients. Singhi & Kumari (2024) examine step-count sensitivity in IG variants and the Important Direction Gradient Integration (IDGI) framework. Their reported stability experiment compares JPEG-compressed and uncompressed inputs. Our diagnostics hold the context fixed and examine repeatability separately from coordinate

changes across budgets. The smooth construction in Appendix B shows a limit of these checks. Agreement on finitely many grids and exact scalar completeness do not establish accurate coordinate integrals.

Petsiuk et al. (2018) introduced the insertion and deletion curves that we use as ranking-based faithfulness metrics. Several studies examine the reliability of perturbation metrics. Hooker et al. (2019) evaluate attributions by retraining on ablated inputs. Rong et al. (2022) show that deletion without retraining leaks information through the shape of the mask itself. Tomsett et al. (2020) also find that common saliency metrics are unreliable. This literature already establishes that evaluation depends on the metric. Our case study examines that dependence for action-discrepancy responses by relating their normalized curves algebraically and rescoring preserved interventions while holding the feature ordering fixed.

Hama et al. (2023) analyze insertion and deletion scores through an anchored decomposition of the response. Their Section 3.3 derives random-order expectations and shows that interactions can move the expected curve area away from its straight-line reference. Their Section 3.4 also gives a case in which Shapley ordering does not maximize AUC. Appendix B.2 gives sufficient conditions for optimal ordering under increasing transformations of additive scores on the intervention vertices. Its IG argument uses a linear score along the continuous path, whose gradients remain proportional to a fixed coefficient vector. Our nonlinear construction allows the coordinate gradient proportions to vary along that path. We examine the specific quadratic and stabilized negative-norm responses, their finite-stabilizer rank reversal and their overshoot behavior in saved RDT interventions. The finite-grid calibration example illustrates the established random-reference concern in this setting.

Adebayo et al. (2018) proposed model and data randomization tests to assess whether saliency maps depend on learned parameters and training data. Their results show why visually plausible maps do not establish this dependence. Our backbone randomizations are limited because the learned encoders and adaptors remain in place. For diffusion generative models in the image domain, cross-attention maps relate generated pixels to prompt tokens (Tang et al., 2023). This differs from attributing the scalar discrepancy of a predicted control chunk.

Our implementation applies IG to a fixed-noise action-discrepancy function with modality-specific baselines. Path weighting is already used in attribution methods such as IDG (Walker et al., 2024). The factor derived here follows specifically from changing the scalar discrepancy target. We apply these established concerns to quadratic and stabilized negative-norm action discrepancies and examine their separate effects on fixed interventions and integrated rankings.

2.2 Vision-Language-Action Models

VLA models combine perception, language understanding and action generation. RT-2 (Zitkovich et al., 2023) fine-tunes a vision-language model to produce discretized actions as token sequences. Octo (Octo Model Team et al., 2024) trains a generalist policy on a large multi-robot dataset using a transformer backbone. Diffusion Policy (Chi et al., 2023) treats action generation as conditional denoising diffusion (Ho et al., 2020). It produces action chunks through iterative refinement rather than single-step regression. RDT (Liu et al., 2025) scales this approach using a Diffusion Transformer (DiT) backbone, a SigLIP vision encoder (Zhai et al., 2023), and a T5-XXL language encoder (Raffel et al., 2020). RDT processes six camera slots, consisting of three camera views over a two-frame history, through SigLIP to produce 4,374 vision tokens. It combines these with up to 1,024 language tokens and a 128-dimensional unified state vector. It then generates 64-step action chunks through a five-step DPM-Solver++ (Lu et al., 2025) denoising schedule. The combination of modalities and repeated denoising steps makes RDT an interesting but challenging model for attribution.

2.3 Interpretability in Robotics

Earlier work on explaining robotic policies focused on reinforcement learning agents in discrete-action settings. Greydanus et al. (2018) applied perturbation-based saliency to Atari agents. Counterfactual methods (Amir-Tai et al., 2024) explain policies by comparing the outcomes of executed and alternative actions. In continuous control, VisualBackProp (Bojarski et al., 2018) visualizes convolutional neural network (CNN) saliency for

end-to-end driving policies. These are convolutional driving models rather than multimodal diffusion samplers. Recent VLA studies examine different quantities to explain. Zhang et al. (2026) estimate visual influence using randomized perturbations and action-discrepancy responses, with explicit assumptions that relate mean-squared error to distributional divergence. VLA-Trace (Shi et al., 2026) combines representation comparisons, attention interventions and behavioral probes. IMPACT-VLA (Kim & Park, 2026) attributes phase-modality interventions to downstream task success through counterfactual closed-loop execution. Our estimand is the discrepancy of a fixed-noise predicted action chunk at a fixed context. It does not explain eventual task success. We examine how response geometry affects the evaluation of IG rankings for this estimand and do not claim to originate VLA interpretability.

3 Attribution Targets and Evaluation

3.1 How Integrated Gradients Is Computed

We start with a differentiable function $F: \mathbb{R}^n \rightarrow \mathbb{R}$, an input $x \in \mathbb{R}^n$, and a baseline $x' \in \mathbb{R}^n$. Integrated Gradients measures each input dimension’s contribution by following the straight path from the baseline to the input. For dimension i , this gives

$$\text{IG}_i(x, x') = (x_i - x'_i) \int_0^1 \frac{\partial F}{\partial x_i} \Big|_{x' + \alpha(x - x')} d\alpha. \quad (1)$$

IG satisfies *completeness* when the fundamental theorem of calculus and the chain rule apply along this path. This means that $\sum_i \text{IG}_i(x, x') = F(x) - F(x')$. We approximate the integral using a uniform Riemann sum with m steps, which gives

$$\text{IG}_i \approx \frac{x_i - x'_i}{m+1} \sum_{r=0}^m \frac{\partial F}{\partial x_i} \Big|_{x' + \frac{r}{m}(x - x')}. \quad (2)$$

The historical implementation gives equal weight to every sampled point, including both endpoints. Convergence depends on the regularity of the function being integrated along the path. At the same time, adding values in finite precision creates another source of error. Increasing m does not separate this error from integration error and does not necessarily reduce it.

3.2 Keeping the Target Fixed Through Denoising

At policy call t , we write the sampler output as $g_\theta(o_t, z_t)$, where o_t is the observation and z_t is the fixed diffusion noise. We call this predicted action chunk μ_t to keep the notation consistent with the historical records. It is a sample produced with fixed noise. The notation does not establish that it is a conditional mean. We compute the reference chunk $a_t = g_\theta(o_t, z_t)$ once without gradients. For an input changed by an intervention, the main attribution target is the auxiliary score

$$Q_t = -\frac{1}{2\sigma^2 D} \|a_t - \mu_t\|^2, \quad (3)$$

where $\sigma^2 = 1$ and $D = 512$ active entries cover 64 predicted time steps over eight active dimensions. This score is the part of a Gaussian log kernel that changes with the action difference, averaged over entries. The log-normalizing constant is left out. It has not been derived as the diffusion policy’s conditional log likelihood. The historical code and files call this target `logpi`. We use ΔQ for its change. Attribution covers the complete predicted chunk, while the execution loop applies a subsampled prefix before replanning or stopping.

We use the same noise realization at every point on the interpolation path. This keeps the scalar function deterministic throughout the calculation. Explaining an expectation over noise would measure a different quantity. It would require justification for interchanging the gradient and expectation, along with a Monte Carlo treatment. Drawing independent noise at every path point does not compute the fixed-noise quantity studied here. The historical implementation replaced nonfinite gradient entries with zero. This can change individual coordinates even when their errors cancel in the sum. As a result, the scalar completeness

residual can be small or even zero while the coordinates remain wrong. Completeness alone cannot establish coordinate accuracy.

3.3 Other Scalar Targets

We compare the quadratic target with two other ways to measure the action difference,

$$y_t^{\ell_2} = -\sqrt{\|a_t - \mu_t\|_2^2 + \epsilon}, \quad y_t^{\max} = -\max_j |a_{t,j} - \mu_{t,j}|, \quad (4)$$

where $\epsilon = 10^{-12}$ is the stabilizer used in the implementation. Both targets use the same D active entries. The stabilized ℓ_2 target is smooth in the residual and has zero gradient when the actions agree exactly. Away from the stabilizer scale, its gradient in action space approaches unit norm. The maximum-deviation target, called maxdev, uses a different norm. It is not generally a monotone transform of ℓ_2 . For example, the residuals (1,1) and (1.1,0) reverse order under these norms. For a fixed perturbation, the quadratic and stabilized ℓ_2 responses have an algebraic relationship. However, their gradients can still produce different attribution rankings. Their units and scales also differ, so a larger magnitude alone does not show that a target provides more information or is more useful. Section 5.1 separates the response changes for the quadratic and stabilized ℓ_2 targets. The maximum-deviation records remain an additional diagnostic in the released analysis.

3.4 How the Response Changes the Weight Along the Path

We hold the reference action, active coordinates, model, noise and other modalities fixed. We then write $r(x) = \|a - \mu(x)\|_2^2$ and $L(x) = -\sqrt{r(x) + \epsilon}$. Where the gradients exist, the chain rule gives the exact relationship

$$\nabla Q(x) = \frac{\sqrt{r(x) + \epsilon}}{D\sigma^2} \nabla L(x). \quad (5)$$

This means that the nonzero gradients point in the same direction at each point. Both gradients are zero when the prediction exactly matches its own reference. Along the path $x(\alpha) = x' + \alpha(x - x')$, the integrated gradient becomes

$$\text{IG}_i^Q = (x_i - x'_i) \int_0^1 \frac{\sqrt{r(x(\alpha)) + \epsilon}}{D\sigma^2} \partial_i L(x(\alpha)) d\alpha. \quad (6)$$

The positive weight generally changes with α . Different coordinates can contribute at different parts of the path. Changing the weight on those parts can therefore change the final ranking after integration. This is consistent with IG implementation invariance, which compares different implementations of the same scalar function (Sundararajan et al., 2017).

For a concrete example, take the scalar action $\mu(x_1, x_2) = x_1^2 + \frac{4}{5}x_2$, baseline (0,0), actual input (1,1), detached reference $a = \frac{9}{5}$, and $D = \sigma^2 = 1$. Exact integration along the same straight path gives

$$\text{IG}^Q = \left(\frac{23}{30}, \frac{64}{75} \right), \quad \lim_{\epsilon \downarrow 0} \text{IG}^L = \left(1, \frac{4}{5} \right). \quad (7)$$

Quadratic IG puts feature 2 first. The stabilized L target with the implemented $\epsilon = 10^{-12}$ puts feature 1 first, with a margin of at least 0.1999975. Appendix A proves this bound with the finite stabilizer. The reversal therefore does not depend on replacing the implemented response with its limiting value. This example shows that a reversal can occur in a smooth nonlinear policy. It does not show that rankings always change or that this effect occurs empirically in RDT.

3.5 Attribution at Each Policy Call

RDT outputs chunks of 64 action steps, which are subsampled $4\times$ to 16 environment steps. We run IG once per *policy call* because the chunk is produced as one decision. We therefore attribute one policy query at a

Algorithm 1 IG at each policy call with fixed noise. Section 4 explains which historical artifacts are available and their limits.

Input. Policy π_θ , environment $\mathcal{E}$, integration steps m , target $y \in \{Q, \ell_2, \text{maxdev}\}$

Output. Attributions $\{A_t^V, A_t^L, A_t^S\}$ at every policy call.

```

1:  $o_0 \leftarrow \mathcal{E}.\text{reset}()$ 
2: for policy call  $t = 0, 1, \dots$  do
3:   Encode  $o_t$  into vision  $v_t$ , language  $l_t$ , state  $s_t$ 
4:   Draw and store  $z_t$ . Compute  $a_t \leftarrow g_\theta(o_t, z_t)$  without gradients
5:   Define  $F$  from  $a_t$ , the fixed-noise predicted chunk  $\mu_t$ , and the chosen target
6:   for modality  $M \in \{V, L, S\}$  do
7:     Set baseline  $x'_M$ , input  $x_M$ 
8:     Freeze the other two modalities
9:      $A_t^M \leftarrow \text{IG}(F_M, x_M, x'_M, m)$ 
10:  end for
11:  Record  $A_t^V, A_t^L, A_t^S$  and completeness errors
12:  Apply action chunk and advance  $\mathcal{E}$  by up to 16 steps, stopping on termination
13: end for
```

Eq. 2

time rather than each environment step. Runtime depends on the implementation, hardware, precision and integration budget. The earlier timing summaries are not used as a performance result.

At each policy call, we compute attribution separately for each modality using its own forward function.

- **Vision.** We keep language and state fixed and interpolate the adapted image tokens from the baseline to the real observation.
- **Language.** We keep vision and state fixed and interpolate the adapted language tokens from a minimal two-token baseline, encoded through T5-XXL, to the real instruction embedding.
- **State.** We keep vision and language fixed and interpolate the 128-dimensional unified state vector from zeros to its real value. For the Franka arm, eight dimensions are active. These represent seven joints and the gripper. The state-adaptor multilayer perceptron (MLP) stays inside the forward function so that the calculation preserves the structure of each joint.

Algorithm 1 shows this procedure.

Choosing the baseline. The vision baseline is a uniform gray image in the external-camera slot. The other five camera slots contain constant background images and remain fixed at their real values. This means that IG attributes only to the slot that contains information about the scene. The language baseline is a minimal two-token sequence made from the tokenizer’s padding and end-of-sequence tokens. T5 has no beginning-of-sequence token. We encode this sequence through T5-XXL instead of using a zero vector. We then add zeros to the encoder output until it reaches the stored embedding length. The language forward passes attend over the real token span for the task. As a result, the attended baseline contains the encoded token pair followed by zero vectors at the remaining attended positions. The GELU-based language adaptor can accept a zero vector numerically, but that would define a different representation baseline. The preliminary comparisons of effectiveness cannot be independently reconstructed. The state baseline is a zero vector in normalized joint space. This choice does not establish that it represents a physically supported state or a useful reference for an explanation. Appendix C.2 reports the available checks of sensitivity to the vision baseline.

3.6 How We Evaluate the Attributions

Completeness. For each modality, we keep the other inputs fixed and report the absolute residual $|\sum_i \text{IG}_i - (F(x) - F(x'))|$ and its relative version

$$\text{Err} = \frac{|\sum_i \text{IG}_i - (F(x) - F(x'))|}{|F(x) - F(x')|}. \quad (8)$$

The relative quantity is undefined when the signed gap is exactly zero. A small nonzero gap can make numerical error appear large after division. The historical action-norm filter $\|a_t\| \geq 15$ does not place a bound on this denominator. We retain it only to check how changing the included population affects the results. A low residual shows agreement in a scalar accounting identity. Errors in individual coordinates can still cancel in the sum while changing the attribution map or ranking.

Perturbation curves. We rank eligible feature groups by adding the absolute coordinate attributions within each group. For vision, the groups are patch tokens from the external camera. For language, they are the attended instruction tokens. Deletion replaces the first groups in the ranking with their baseline features. Insertion starts from the baseline input and replaces the first groups with their original features. For a scalar response R , the normalized value at each point is

$$\tilde{R}(x_k) = \frac{R(x_k) - R(x')}{R(x) - R(x')}. \quad (9)$$

We integrate using the fraction grid stored in each record. The curve and its AUC can both fall outside $[0, 1]$. The historical grid requests $\{0, 1, 5, 10, 20, 30, 50, 75, 100\}\%$ of the groups. Rounding changes the actual fractions of language tokens. The saved curves therefore cannot be treated as continuous functions sampled at uniform intervals. Top- k score changes are $R(x_k) - R(x)$, with units determined by R . We keep the attribution target that produces the ranking separate from the response that scores its interventions. The revised tables do not use the earlier study-specific thresholds as tests of effectiveness.

Randomization and baseline checks. C1 reinitializes model parameters and measures the correlation between the resulting attributions and the originals. The standard version changes the final action-output projection and recomputes its reference action. The frozen-reference and backbone-cascade versions keep the original reference. The historical cascade reinitializes all backbone parameter matrices of rank at least two at the same time. It keeps one-dimensional parameters, encoders and adaptors. It therefore does not progressively randomize one layer at a time. C2 shuffles both the external-camera pixels and the language. The historical implementation uses a pre-T5 shuffled-language cache when one is available. Otherwise, it permutes the post-T5 embeddings. The saved records do not establish which mode was used for each call. These tests measure different forms of dependence. C2 is not the label-randomization experiment of Adebayo et al. (2018). Historical perturbed maps use $m = 16$, while original maps nominally use $m = 64$. C1 therefore changes the numerical approximation as well as the model weights. We also compare black, gray and blurred observation features to check baseline sensitivity.

4 Retrospective Data and Statistical Protocol

Scope and traceability. The revised analysis checks the bytes of 99 preserved files containing 22,966 valid records and 1 specifically identified corrupt line. Each valid physical record is identified by its source path, original line number and byte hash. Some recorded keys occur more than once. The inventory contains 129 such groups, and 58 of them differ in scientific fields as well as, or instead of, timing metadata. We use the last occurrence as a declared rule for this retrospective analysis. This rule does not establish that the selected attempt is correct when the records conflict. To show how those conflicts affect the results, a separate sensitivity analysis removes every key with a conflicting payload. The released lineage registry links the records and occurrence decisions to population membership, result identifiers and generated manuscript cells. These links allow the preserved bytes to be checked. They cannot recover missing checkpoints, observations, action tensors or noise.

Campaigns and models. The main reconstructed RDT-170M tables use the saved Month 7 verification campaign at $m = 64$. That campaign covers PickCube, StackCube, PegInsertionSide and PickSingleYCB in ManiSkill3 (Tao et al., 2024). These experiments are separate from the original main pass, whose inputs are unavailable. The verification campaign used pretrained RDT-170M, while the saved RDT-1B PickCube campaign used a task-finetuned checkpoint. The comparison therefore changes training, trajectories and historical preprocessing along with parameter count. It cannot isolate the effect of model scale or competence. For this reason, Appendix C reports the 1B cohort separately. The primary tables do not include missing original completeness or standard-sanity results, or missing early PickCube/StackCube integration-budget results.

Known protocol limits. The historical attribution covers the full predicted action chunk, although the simulator executes a subsampled chunk. The choice of four tasks also reflects historical implementation decisions. An attempted TurnFaucet run used an incorrect controller, and no corrected rerun is documented. Its unsuccessful outcomes cannot establish that the model lacked the capability for that task. Early follow-up jobs replaced encoded padding/end tokens with a zero language baseline. The PickSingleYCB follow-up also used a proxy PickCube instruction embedding. As a result, the pooled 1B language cohort contains both the early baseline and later encoded baselines. Early sidecar filenames omitted the seed and integration budget. This allowed files to be overwritten and possibly reused across nominal runs. The seed-tagged Month 7 records reduce that filename problem, but they do not recover the old contexts. Matching scalar fingerprints alone therefore do not establish exact context equality.

Estimands and uncertainty. The primary population contains all valid saved calls after applying the declared occurrence rule. We summarize continuous call-level quantities by their medians over calls. Residual pass rates are proportions over the stated call population. Solver contrasts report relative changes between endpoint medians, while the separately named equal-episode analysis uses episode medians. For uncertainty, we resample whole recorded episode groups, identified by task, model, recorded seed and episode, with $B = 10,000$ draws and seed zero. Each point estimate and its interval use exactly the same source-record population. The resulting percentile intervals describe the released empirical mixture of episodes. Longer episodes contribute more calls to that mixture, and the intervals do not measure variability across independently trained checkpoints. A separately named 1B sensitivity gives each episode equal weight by reporting the median of episode medians. For the old randomization variants, we also group nominal task/episode keys across seeds when they may have been reused. This is a conditional sensitivity to dependence between those records. Neither grouping can recover unobserved context identity. We report very small cluster counts explicitly. In particular, resampling two baseline groups is not evidence of calibrated interval coverage. All unfiltered results and the historical norm-selected sensitivity remain available. We do not remove a tail to obtain a favorable threshold verdict.

The additional paired rescoring analysis holds each call’s ranking and interventions fixed. It defines $\Delta_{ej} = \text{AUC}_{Q,ej} - \text{AUC}_{L,ej}$ within call j of recorded episode e . Its primary estimand is $E^{-1} \sum_{e=1}^E n_e^{-1} \sum_{j=1}^{n_e} \Delta_{ej}$, which gives episodes equal weight. The median of paired call differences is a prespecified descriptive sensitivity. Both summarize changes within calls, so neither is the difference between the marginal medians reported above. Before computing these summaries, the protocol specified all eight ranking/modality/direction combinations and the same 10,000-draw episode-bootstrap convention. The intervals are conditional descriptions without a multiplicity adjustment. We do not claim calibrated joint coverage or global significance. If either response has a zero endpoint gap, the paired difference remains undefined and stays in the accounting. All 750 calls in each ranking cohort have defined paired differences. Each of the 30 recorded episodes contains 25 calls, so the equal-episode and equal-call means coincide here.

5 Response Geometry and Retrospective Results

5.1 Separating Attribution Target from Evaluation Response

We can change the response used to score a recorded intervention without recomputing IG or changing any selected feature. Let d be the recorded active-action discrepancy, $Q = -d^2/1024$, and $L = -\sqrt{d^2 + \epsilon}$ with

$\epsilon = 10^{-12}$. The responses are related by

$$L = -\sqrt{-1024Q + \epsilon}, \quad Q = -\frac{L^2 - \epsilon}{1024}. \quad (10)$$

When the actual endpoint is an exact self-reference and the all-baseline squared residual is $R > 0$, their normalized values satisfy

$$\begin{aligned} N_Q(r) &= 1 - \frac{r}{R}, & N_L(r) &= \frac{\sqrt{R + \epsilon} - \sqrt{r + \epsilon}}{\sqrt{R + \epsilon} - \sqrt{\epsilon}}, \\ N_Q(r) - N_L(r) &= \frac{(\sqrt{r + \epsilon} - \sqrt{\epsilon})(\sqrt{R + \epsilon} - \sqrt{r + \epsilon})}{R}. \end{aligned} \quad (11)$$

Thus $N_Q > N_L$ for $0 < r < R$. The responses are equal at both endpoints, and their order reverses for $r > R$. If every intervention stays in range, integration with nonnegative weights preserves this weak ordering even when the curve is not monotone. The area inequality is strict only if an interior residual receives positive integration weight. Overshoots can reverse the area ordering, so they must be retained rather than clipped. If $R = 0$, neither normalization is defined. For an in-range curve, changing from Q to L therefore lowers both insertion and deletion AUC. Larger insertion AUC and smaller deletion AUC are often interpreted as better ranking quality. Under that interpretation, this response change points to opposite changes in quality even though the interventions stay the same. The historical computation does not assume that the recorded actual endpoint is an exact self-reference. Instead, we transform every saved curve ordinate, compute its endpoints under the transformed response, and integrate again on the same saved grid. After duplicate handling, each ranking cohort contains 750 calls from 30 episodes. For quadratic-target vision rankings, the median deletion AUC changes from 0.4479 under Q to 0.2908 under L . For stabilized- ℓ_2 rankings, it changes from 0.2830 under L to 0.4404 under Q . Table 1 and Figure 1 show both modalities and both perturbation directions.

Table 1: The response is changed while each recorded ranking stays fixed. The table shows median AUC with conditional 95% episode-bootstrap intervals. Each ranking cohort contains 750 calls from 30 episodes and reuses the same recorded interventions. The two ranking cohorts have not been established as paired contexts. Comparing them therefore cannot show that one ranking is better.

Ranking target	Response	Modality	Insertion AUC	Deletion AUC
Q	Q	Vision	0.883 [0.845, 0.909]	0.448 [0.408, 0.504]
Q	Q	Language	0.580 [0.510, 0.658]	0.372 [0.311, 0.428]
Q	L	Vision	0.756 [0.718, 0.785]	0.291 [0.253, 0.333]
Q	L	Language	0.469 [0.412, 0.516]	0.257 [0.228, 0.284]
L	Q	Vision	0.884 [0.833, 0.921]	0.440 [0.402, 0.501]
L	Q	Language	0.635 [0.582, 0.720]	0.368 [0.302, 0.407]
L	L	Vision	0.756 [0.700, 0.803]	0.283 [0.252, 0.327]
L	L	Language	0.507 [0.456, 0.566]	0.249 [0.202, 0.278]

These results show that scores within each saved ranking cohort depend on response geometry. They do not show that Q-IG and L2-IG select the same features, produce equivalent rankings, or provide effective rankings. The attempted join between cohorts found different reference-action norms at every nominal key. Exact raw contexts and noise are also unavailable. Apparent agreement after common rescaling therefore cannot take the place of a prospective paired comparison of rankings. A larger native L2 score likewise does not by itself represent recovered information. Stabilized L2 has no square-root kink at exact agreement.

Paired changes and overshooting tails. Table 2 shows why the choice of summary matters. All eight paired-call medians are positive, ranging from 0.1041 to 0.1590. At the same time, six equal-episode means are negative. For Q-ranking vision insertion, the median paired change is 0.1042, while the mean is -17.0966 with conditional interval $[-33.8210, -3.9746]$. The most negative paired change in that case is

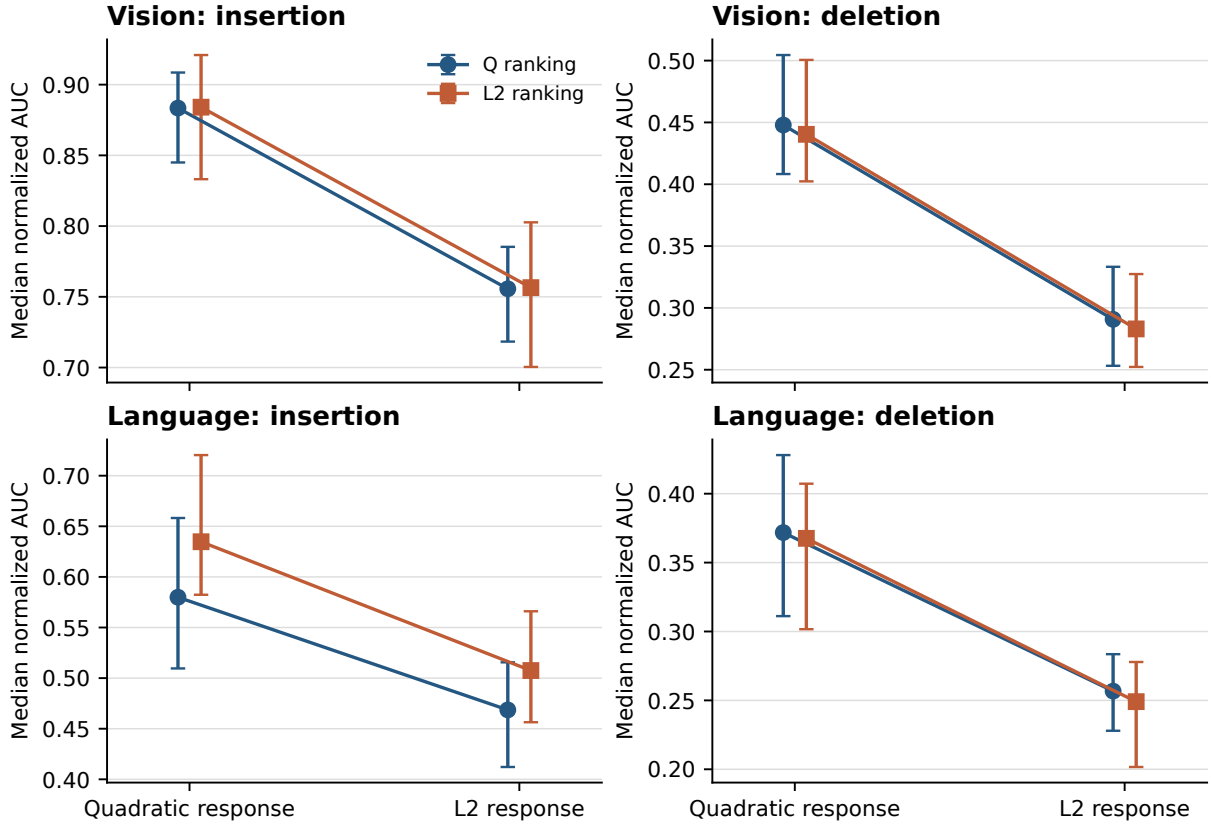


Figure 1: Changing the evaluation response while holding each recorded ranking and its interventions fixed. Error bars give conditional episode-bootstrap intervals for each median. Each line connects responses within one ranking cohort. The two ranking cohorts are not authenticated pairs, and there is no universal chance reference line.

−4438.8386. These finite extremes remain in the analysis without clipping or trimming. Appendix D reports an exploratory influence check for all eight cases. All four negative insertion means stay negative when any one recorded episode is omitted. The two negative language-deletion means can change sign under these omissions, which is consistent with their conditional intervals already crossing zero. This check shows how the results are concentrated in the recorded cohort. It does not establish population robustness or replace the untrimmed primary estimands.

Table 2: Paired response differences $\Delta = \text{AUC}_Q - \text{AUC}_L$ within the same saved curve. Each row uses 750 calls from 30 recorded episodes. Rows within a ranking cohort reuse those calls. The primary mean gives episodes equal weight, and its 95% interval is a conditional, unadjusted episode-bootstrap description. The paired-call median is also shown because overshooting tails can reverse the mean’s sign. Neither summary compares the quality of Q and L2 rankings.

Ranking	Modality	Curve	Mean Δ	95% interval	Median Δ
Q	vision	insertion	-17.0966	[-33.8210, -3.9746]	0.1042
Q	vision	deletion	0.1227	[0.1059, 0.1382]	0.1588
Q	lang	insertion	-0.1819	[-0.4051, -0.0277]	0.1169
Q	lang	deletion	-0.0095	[-0.1022, 0.0631]	0.1189
L2	vision	insertion	-18.5620	[-37.8672, -3.1731]	0.1041
L2	vision	deletion	0.0184	[-0.2037, 0.1378]	0.1590
L2	lang	insertion	-0.0329	[-0.0963, 0.0266]	0.1279
L2	lang	deletion	-1.3710	[-4.2081, 0.0780]	0.1205

To examine those tails, we classify each saved squared-discrepancy value relative to the interval between its stored actual and baseline endpoint discrepancies. This classification does not assume exact self-reference. A high-distance overshoot exceeds both endpoints. A low-distance undershoot falls below both. The artifact records which endpoint is larger and retains both kinds of departure, undefined gaps and a prespecified sensitivity to boundary labels. In Q-ranking vision insertion, 312 of 750 calls contain at least one high-distance overshoot, and 89 have a negative paired AUC change. Breaking down the trapezoidal difference by saved-node category gives mean contributions of 0.1144 from in-range nodes and −17.2111 from high-distance overshoots. There is no low-distance contribution in that case. These contributions sum to the reported mean. They show that the negative contribution from the recorded overshoots is large enough to reverse its sign. The L2-ranking vision-insertion cohort has the same qualitative disagreement between its mean and median in its own records. Node categories and segment endpoint labels describe only sampled interventions. They do not establish how the continuous curve behaves between those points. The in-range ordering in Equation 11 therefore does not imply that a response change moves every curve in an empirical population in the same direction.

5.2 An Analytic Random-Order Calibration Example

Random-order AUC need not match a straight-line reference when the scored response contains interactions (Hama et al., 2023). The following example gives the exact areas for our quadratic response and finite evaluation grids. Consider the finite additive scalar policy $f(x) = \sum_{i=1}^n x_i$, with an all-one input, a zero baseline and the self-reference $f(x) = n$. Every feature ordering produces the same perturbation values. Interventions on whole-feature prefixes realize only the fractions $q = k/n$, with $k = 0, \dots, n$. At those fractions, the normalized quadratic deletion and insertion values are $1 - q^2$ and $1 - (1 - q)^2$. Trapezoidal integration over all $n + 1$ prefix fractions gives

$$\text{AUC}_n = \frac{2}{3} - \frac{1}{6n^2}$$

for either curve. Integrating the polynomial extensions over continuous $q \in [0, 1]$ instead gives $2/3$. This is also the limit as $n \rightarrow \infty$, but it is not the exact finite-grid area. With $n = 100$, every fraction on the historical nine-point requested grid is realized exactly. Trapezoidal integration on that grid gives 0.65976 for both curves and every ordering. The policy is additive, but its quadratic discrepancy response contains interactions between features. This finite example illustrates why a universal random-order AUC of 0.5 does

not apply to that response. For comparison, an unstabilized negative-norm response is linear on the same fractions and has trapezoidal AUC 0.5. A fixed AUC reference such as 0.5 must therefore be justified for the particular policy, response and intervention protocol. This construction is an analytic control. It is not an empirical measurement of random-ranking performance on RDT.

5.3 Completeness and Perturbation Summaries

Table 3 reports completeness residuals from the saved verification campaign. Vision and language have smaller median relative residuals than state in these cohorts. This does not establish that their coordinates are more accurate or that dimension count causes the difference. The historical integrator combines errors from the model forward pass, path arithmetic, summation and quadrature. Before choosing production settings, actual-model validation must separate these factors using identical inputs.

Table 3: Results from the Month 7 verification campaign using all valid calls at $m = 64$. Each cell shows the median relative completeness residual as a percentage, with a conditional 95% episode-bootstrap interval. The estimate and interval use exactly the same calls and episodes. These results describe historical numerical checks. They do not establish that the individual attribution coordinates are accurate.

Task	Calls	Episodes	Vision (%)	Language (%)	State (%)
PegInsertion	700	100	2.17 [1.94, 2.35]	1.83 [1.69, 1.98]	3.48 [3.00, 4.04]
PickCube	397	100	2.20 [1.90, 2.43]	1.88 [1.62, 2.30]	5.88 [4.99, 7.29]
PickYCB	397	100	1.85 [1.71, 2.06]	2.02 [1.79, 2.18]	4.26 [3.50, 5.62]
StackCube	400	100	2.39 [2.18, 2.59]	1.76 [1.58, 1.97]	10.53 [8.07, 13.13]

Table 4 reports native quadratic-response insertion and deletion AUCs for the same verification campaign. These scores describe the recorded perturbations. Without matched random and simple ranking controls, their magnitudes do not establish attribution efficacy. Normalization can also produce extreme negative values. The smallest saved PickCube vision insertion AUC in the all-valid population is -18190.36. We retain that observation and its endpoint denominator. Neither clipping the value nor applying the action-norm filter would establish its validity. The released denominator and tail diagnostics therefore accompany the medians, since a central summary cannot describe the whole curve population.

Table 4: Results from the Month 7 verification campaign using all valid calls and the native quadratic response. Each median and its conditional 95% interval use the same population. We do not call an entry successful based on a chance threshold that has not been calibrated. The stored token-fraction grids, endpoint conventions and limits on the recorded provenance still apply to these descriptive estimates.

Task	Modality	Calls	Episodes	Insertion AUC	Deletion AUC
PegInsertion	Vision	700	100	0.731 [0.688, 0.774]	0.280 [0.234, 0.309]
PegInsertion	Language	700	100	0.320 [0.257, 0.379]	0.341 [0.291, 0.382]
PickCube	Vision	397	100	0.822 [0.771, 0.838]	0.167 [0.137, 0.213]
PickCube	Language	397	100	0.525 [0.468, 0.580]	0.433 [0.391, 0.475]
PickYCB	Vision	397	100	0.844 [0.808, 0.863]	0.337 [0.296, 0.372]
PickYCB	Language	397	100	0.335 [0.227, 0.409]	0.471 [0.438, 0.500]
StackCube	Vision	400	100	0.821 [0.777, 0.849]	0.208 [0.161, 0.261]
StackCube	Language	400	100	0.359 [0.292, 0.432]	0.480 [0.429, 0.512]

6 Numerical Diagnostics from an Incomplete Campaign

The later diagnostic campaign tests identified current computations, separately from the historical records above. Historical rescoreing applies arithmetic to saved curves whose original contexts cannot be recovered.

The later probe instead evaluates a separately identified offline fp32 function on newly authenticated contexts, using its own reference actions. It cannot validate the historical attribution maps or provide a paired comparison with them. The campaign stopped because of compute cost before its planned numerical coverage and independent verification were complete. We analyze a fixed, authenticated snapshot of sealed results. Entries that were unfinished at that point are not classified as successful or failed calculations. The stopping decision gives no statistical sampling rule for the completed subset. Counts of passed or violated criteria therefore describe that subset, rather than estimate population failure rates.

Functions and precision controls. Stored observations, masks, reference actions and exact noise arrays allow comparisons on fixed, authenticated inputs. The diagnostic implementation separates accumulation arithmetic, master-path and final-product arithmetic, quadrature, and downstream model precision. It compares full coordinate arrays, ranks over eligible groups, selected groups and perturbation responses. An earlier downstream-fp32 factor control lifts the same bf16-rounded parameters and cached adapted representations. It also explicitly converts non-parameter dtype attributes and keeps the original reference action fixed. This changes downstream arithmetic while keeping the numeric inputs fixed. It does not recover unrounded checkpoint parameters or test fp32 encoders. Keeping the original reference also makes this control different from the later probe. The earlier comparisons came before the full repeated-gradient checks, so they cannot alone establish a globally repeatable precision contrast.

The later engineering campaign defines a separate deterministic fp32 function. Image and language representations keep their cached bf16 values lifted to fp32, raw state is re-adapted in fp32, and the function generates its own fp32 action reference. It uses contexts visited by the bf16 policy and the stored source noise. This is an offline intervention function. It is neither the bf16 behavior policy nor a derivative of that policy’s executed commands. Repeating its endpoints checks a different property from repeating its path gradients and attribution maps. Changing the function or its reference therefore cannot be interpreted as changing quadrature alone.

Engineering design and interpretation. The predeclared engineering bank covers four RDT-170M tasks and two tasks using an independently identified authors’ 1B checkpoint. That identity does not recover or authenticate the checkpoint used for the historical task-finetuned 1B cohort in Appendix C. Source collection allowed at most 13 policy calls, with a further cap of 400 environment steps and possible earlier termination. Call indices 0 and 12 were chosen in advance. These are engineering contexts at fixed call indices, rather than uniformly sampled calls from complete trajectories. Collection did not choose episodes by success or action norm. The integration ladders use trapezoidal quadrature with fp32 path and accumulation arithmetic. The same gradient stream also gives a mean path-gradient map without the IG input-difference multiplier. This map is part of the numerical checks, not an evaluated efficacy baseline. The protocol compares coordinates, ranks, completeness and common perturbation responses across integration budgets and repeated calculations. Its tolerances diagnose changes in the tested approximation. They are neither efficacy margins nor calibrated bounds on error relative to an exact attribution. The finest available map is therefore a comparator, not ground truth. The preserved cohort analysis includes every planned context, including calls that the source rollout did not reach and entries with no sealed result.

Preserved coverage. The retained snapshot is labeled October 1, 2026, 03:36 UTC and precedes the later compute stop. Table 5 includes all 24 planned contexts and 144 target/modality cells. Twelve contexts have all six diagnostics complete. They come from six episodes across the four 170M tasks and account for 72 complete cells. Three planned calls were not reached by their source rollouts, accounting for 18 cells. The other nine available contexts, or 54 cells, have no sealed diagnostics at this cutoff. No 1B numerical context is complete in this snapshot. These statuses describe the snapshot. They do not assert the final state of work that was later interrupted.

Repeatability and budget sensitivity. The image and language ladders use $m = 128, 256, 512$, while the state ladder uses $m = 256, 512, 1024$. The finest maps were repeated twice, and seven fixed path locations were probed three times each. All recorded repeated-condition equality and coverage checks met their conditions. These checks depend on the same finite set of contexts. They are not independent replications and do not estimate a population reliability rate.

Table 5: The full planned engineering roster at the authenticated 03:36 UTC snapshot. The first four numerical columns count contexts, while the last gives complete/planned target-modality cells. Each context has six cells. Calls not reached by the source rollout remain separate from available contexts with unsealed results. A complete entry does not imply numerical qualification.

Stratum	Planned	Complete	Unreached	Unsealed	Cells
PegInsertionSide 170M	4	2	0	2	12/24
PickCube 170M	4	4	0	0	24/24
PickCube 1B	4	0	2	2	0/24
PickSingleYCB 170M	4	2	0	2	12/24
StackCube 170M	4	4	0	0	24/24
StackCube 1B	4	0	1	3	0/24
Total	24	12	3	9	72/144

Table 6 reports every completed target/modality group using its two finest recorded budgets. For maps A_m and A_{2m} , the coordinate discrepancy is $\|A_m - A_{2m}\|_1 / \|A_{2m}\|_1$. It is undefined when the denominator is zero. The predeclared relative-coordinate and completeness thresholds are each 1%. Further checks cover group ordering and sensitivity of the common responses. The table counts a violation when a context/target/modality arm exceeds at least one pairwise threshold in this finer-budget comparison. This count includes the mean path-gradient diagnostics. Finest-budget completeness is summarized separately. All lower-budget comparisons, raw denominators and individual check results remain in the artifact. The reported maxima summarize only these recorded cases, without intervals or population generalization.

Table 6: Diagnostics on all 12 completed contexts per target/modality arm. n counts contexts, and “Viol.” counts arms with at least one recorded pairwise-threshold violation between the two finest budgets. The remaining columns give maxima across those contexts: relative IG coordinate L_1 difference, finest-budget relative completeness residual, and maximum raw RMS response-curve change divided by the all-baseline RMS displacement, all in percent. RMS maxima include IG and mean-path-gradient rankings in both perturbation directions. Coordinate differences use the finer map’s L_1 norm. Comparators are 256 versus 512 for vision/language and 512 versus 1024 for state. These dependent checks neither estimate population error rates nor prove that the finest maps are accurate. No setting or subset is approved.

Modality	Target	n	Viol.	IG L_1 (%)	Residual (%)	RMS (%)
Language	L	12	0	0.402	0.315	0.000
Language	Q	12	0	0.006	0.002	0.000
State	L	12	2	20.778	6.655	18.573
State	Q	12	2	32.119	11.436	1.641
Vision	L	12	4	1.234	0.288	7.361
Vision	Q	12	1	2.049	0.124	0.000

At both recorded PickSingleYCB episode-1 contexts, both targets still violate the state completeness threshold at the finest budget. Their two finest maps also differ substantially. The scalar gaps exceed the declared near-zero cutoffs, and the artifact retains their signed gaps and absolute residuals. These discrepancies therefore cannot be explained away by division by an almost-zero scalar gap. In a state case, the intervention response curves also stay unchanged despite a substantial coordinate change. This shows why stable responses alone cannot qualify the integral. The quadratic target fails as well, so the stabilized negative-norm target’s endpoint behavior cannot explain all the failures. The sparse stored path probes do not identify a single numerical cause.

Repeatability and integration-budget sensitivity can therefore occur together. A deterministic calculation can reproduce the same approximation at one budget even though that approximation changes at another budget. At the same time, coordinate errors can cancel in their sum and leave a small completeness residual.

Neither repeatability nor one completeness statistic is enough to qualify the production setting. Appendix B explains why agreement across several budgets is also insufficient in general. Equal and opposite coordinate errors can preserve exact completeness while all selected coarse grids produce the same ranking and response curves, even though their ordering differs from exact IG. Correctness here means recovering the specified IG integral. The quality of a perturbation ranking is a separate question, and this analytic possibility does not identify the cause of the observed RDT failures. The failures support a numerical diagnostic conclusion within the tested scope. The incomplete campaign does not establish accuracy across modalities, tasks, checkpoints or future contexts.

7 Discussion and Limitations

The saved records let us separate three questions: what a scalar response measures, whether a numerical attribution is reliable, and whether its ranking contains useful information that depends on the model. Rescoring fixed recorded interventions answers the first question because it changes the response while keeping the selected features fixed. This operation alone cannot answer the other two questions. Our claims therefore concern exact relationships and measurements reconstructed from the records, with numerical diagnostics limited to their stated coverage. Establishing ranking efficacy would require a new paired study with common responses and informative controls. The present results do not establish it. The elementary identities identify which operation changes a reported evaluation. This helps prevent a response change from being interpreted as an empirical improvement in ranking, without proposing a new general theory of attribution. The random-reference and attribution-versus-AUC distinctions have direct precedent in Hama et al. (2023). Our contribution concerns the specified action-discrepancy responses, constructive examples and retrospective measurements within the stated limits. A common response defines the quantity being compared. It does not provide a universal measure of explanation quality. Its action coordinates, units, baselines, perturbation groups and aggregation need to be justified for the intended use. Matched random, input-difference and path-gradient controls would help address the unresolved ranking question by separating information in the attribution from simple input structure and the IG multiplier. The deferred protocols provide no comparative or learned-weight-specific result in this paper.

A practical evaluation procedure. These distinctions suggest four steps when reporting action-discrepancy attributions.

1. Specify the scalar attribution target and evaluation response separately. Also state the reference action, fixed noise, feature representation, baseline, intervention groups and schedule.
2. Compare rankings on authenticated matched contexts under each common response. Claims of efficacy need matched random and simple ranking controls. To examine a response change, rescore the same interventions, recompute the endpoint normalization and retain overshoots and undefined gaps.
3. Check repeatability and sensitivity of coordinates, rankings and responses across numerical settings on a declared context roster. Retain failures and incomplete cases. Completeness and agreement on finitely many grids help diagnose numerical behavior, but they do not certify exact IG.
4. Report the unit of analysis and dependence grouping. Distinguish response sensitivity from numerical reliability and ranking quality, because none of these alone establishes improved closed-loop behavior.

This procedure clarifies what each comparison identifies. It does not determine a universally adequate integration budget or evaluation response.

The historical records also contain several sources of uncertainty that require different treatment. Source-record hashes can authenticate the released bytes, but checkpoint and context identities can remain unknown. Episode resampling addresses dependence within episodes conditional on a particular empirical mixture. It cannot repair shared or overwritten contexts, preprocessing confounds or missing training replicates. The

1B results come from one checkpoint. The solver and alternative-target analyses also cover narrower task populations than the verification campaign. Endpoint normalization can produce unbounded response curves, making the results sensitive to endpoint gaps. Camera-token and language-token interventions differ in their group counts and meaning. Their AUCs therefore cannot be used interchangeably as measures of modality importance.

We preserve historical overlay panels as artifacts, but they do not establish successful grounding or diagnose behavioral failure. Groups previously named from completeness residuals likewise describe numerical diagnostics. They are not independently validated labels for outcomes such as misgrounding or delayed credit. The original main-pass and early budget-extension numbers could not be recovered, so the revised primary tables exclude them. Appendix F reports only later budget cohorts that can be reconstructed. Those cohorts do not establish convergence.

8 Conclusion

An explicit auxiliary action-discrepancy score defines per-policy-call IG for a fixed-noise diffusion sampler. For the saved RDT perturbation curves, changing only the evaluation response changes normalized AUC while leaving the selected features fixed. All eight reported paired cases have positive median changes. At the same time, overshooting tails make six means negative, which shows why the summary statistic and endpoint geometry both affect the conclusion. A finite-grid calibration example illustrates the response dependence of the random-ranking reference for the same evaluation protocol. The canonical retrospective tables retain the measurements supported by the available records and exclude primary constants that could not be recovered. The actual-model diagnostics preserve unresolved numerical failures and incomplete coverage. The smooth aliasing construction shows a further limit: finite grids can agree while producing an ordering different from exact IG. A common response is needed to interpret a comparison of ranking quality. Our results do not establish that one target or attribution method performs better under that response.

Ethical Statement

This work analyzes pretrained robotic manipulation policies in simulation. No human subjects were involved. Attribution serves as a diagnostic tool here and does not guarantee safe robot deployment. Attractive maps, low completeness residuals or uncalibrated AUCs can create unjustified confidence when they are misinterpreted. The stated limits and controls are intended to prevent that inference.

Reproducibility Statement

The retrospective analysis parses a raw inventory authenticated by hashes, records decisions about repeated occurrences, and links each result to physical source records. The revised tables and figures come from the same canonical artifact. Their points and intervals use identical populations, and a manuscript lineage registry records their sources. The artifact retains exclusions for conflicting keys, action-norm strata, endpoint and tail diagnostics, alternative episode weightings and conditional grouping sensitivities. The separate paired rescoring artifact includes its pre-computation analysis protocol, all eight cases, paired call and episode summaries, bootstrap membership, and an exact decomposition by saved-node geometry. A separately versioned exploratory supplement records every single-episode omission and its source identities while leaving those primary results unchanged. The smooth nested-grid example includes an analytic proof and repeated FP32/FP64 CPU checks through the unchanged trapezoidal kernel. These operations reproduce arithmetic over saved records. They do not recover missing models, sidecars or original observations. The portable numerical projection retains the full planned roster, recorded checks and metrics, report hashes, and selected YCB FP32 tensors with their tensor hashes. It does not contain all raw artifacts for the 72 completed target-modality cells. Its CPU reproducer uses this projection to recompute the reported diagnostics and saved-tensor checks. It does not repeat the complete original raw-artifact audit or execute the model again. The new runtime separately records immutable context, source, checkpoint, language, environment and noise identities. It fails explicitly when an attribution is nonfinite instead of sanitizing its coordinates. A

candidate PDF or successful script run alone cannot establish an authenticated scientific replication. Anonymous manuscript sources and a research supplement accompany this audit copy. The identified repository URL is omitted to support independent review.

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A Response-Geometry Derivations

The identities in Section 3.4 hold the action reference fixed and assume that the relevant path gradients are integrable. Differentiating $Q = -r/(2D\sigma^2)$ and $L = -\sqrt{r+\epsilon}$ gives Equation 5 directly. For Equation 11, set $a_0 = \sqrt{\epsilon}$, $b = \sqrt{R+\epsilon}$ and $z = \sqrt{r+\epsilon}$. Because $R = b^2 - a_0^2$ and $r = z^2 - a_0^2$, rationalization gives

$$\frac{b^2 - z^2}{R} - \frac{b - z}{b - a_0} = \frac{(z - a_0)(b - z)}{R}. \quad (12)$$

For in-range residuals, the maximum difference is $(b - a_0)^2/(4R) \leq 1/4$. These pointwise statements apply to interventions from a fixed ranking. They do not order different rankings or establish improvements in task success.

For the rank-reversal example, let $c = 4/5$ and parameterize the path by (t, t) . The scalar residual is $d(t) = 1 + c - t^2 - ct = (1 - t)(1 + t + c) \geq 0$. The quadratic path integrands are $2td(t)$ and $cd(t)$, which give

$$\text{IG}^Q = \left(\frac{1}{2} + \frac{c}{3}, \frac{2c}{3} + \frac{c^2}{2} \right) = \left(\frac{23}{30}, \frac{64}{75} \right). \quad (13)$$

For the stabilized negative norm, the corresponding integrands are $2td(t)/\sqrt{d(t)^2 + \epsilon}$ and $cd(t)/\sqrt{d(t)^2 + \epsilon}$. The bound $d(t) \geq c(1 - t)$ and monotonicity of $1 - d/\sqrt{d^2 + \epsilon}$ give

$$\int_0^1 \left(1 - \frac{d(t)}{\sqrt{d(t)^2 + \epsilon}} \right) dt \leq 1 - \frac{\sqrt{c^2 + \epsilon} - \sqrt{\epsilon}}{c} \leq \frac{\sqrt{\epsilon}}{c}. \quad (14)$$

Therefore $0 \leq 1 - \text{IG}_1^L \leq 2\sqrt{\epsilon}/c = 2.5 \times 10^{-6}$ and $0 \leq c - \text{IG}_2^L \leq \sqrt{\epsilon} = 10^{-6}$ at the implemented stabilizer. All coordinates are positive. The stabilized L margin is at least 0.1999975, while Q gives the opposite ordering with exact margin 13/150. The artifact checks these identities through independent differentiation, integration and normalization. Those checks include overshoots, zero gaps and a policy example without rank reversal.

B A Smooth Counterexample to Nested-Grid Agreement

Finitely many integration grids can agree while giving inaccurate coordinate integrals. Exact completeness and unchanged perturbation curves do not remove this limitation. The following construction shows why without a model experiment. All equalities in the construction are statements in real arithmetic.

For a positive integer M , define the smooth scalar policy and target

$$h(t) = \frac{2}{5} \sin^2(\pi Mt), \quad (15)$$

$$\mu(x, y) = \frac{2}{5}x + \frac{3}{5}y + (x - y)h\left(\frac{x + y}{2}\right), \quad (16)$$

$$Q(x, y) = -\frac{1}{2}(\mu(x, y) - 1)^2. \quad (17)$$

Take baseline $(0, 0)$, actual input $(1, 1)$ and the fixed own-reference action $\mu(1, 1) = 1$. This gives the same quadratic self-reference form as the main target, with one active action entry and $\sigma^2 = 1$. Along the straight path (t, t) , the transverse term in μ vanishes. As a result, $\mu(t, t) = t$, $Q(t, t) = -(1 - t)^2/2$, and

$$\nabla Q(t, t) = (1 - t) \left(\frac{2}{5} + h(t), \frac{3}{5} - h(t) \right). \quad (18)$$

The endpoint score gap is $1/2$. Both coordinate integrals are positive, and their exact values differ.

Let T_m denote componentwise composite trapezoidal integration with m intervals. Whenever m divides M , every node $t = k/m$ has $h(t) = 0$. The remaining coordinate functions are linear, so

$$T_m = \left(\frac{1}{5}, \frac{3}{10} \right), \quad \mathbf{1}^\top T_m = \frac{1}{2} = Q(1, 1) - Q(0, 0). \quad (19)$$

Every such grid therefore has zero completeness residual and the same strict absolute-coordinate ranking, which places feature 2 first. Under a fixed deterministic ranking rule, this shared order produces identical insertion and deletion masks. Evaluating those masks on the same policy then gives identical response curves, even when the coordinate integrals are inaccurate.

At the same time, $h(1-t) = h(t)$ and $\int_0^1 h(t) dt = 1/5$, which give

$$\int_0^1 (1-t)h(t) dt = \frac{1}{10}, \quad \text{IG}(Q) = \left(\frac{3}{10}, \frac{1}{5} \right). \quad (20)$$

The exact ranking places feature 1 first. The coarse-grid errors are $(-1/10, 1/10)$. Their sum cancels exactly, but their coordinate L_1 norm is $1/5$, or 40% of the exact map’s L_1 norm. The path score itself is a quadratic polynomial. The unresolved variation is in how its gradient is divided between coordinates, which a scalar completeness check cannot recover.

For this construction, the $2M$ grid resolves the discrepancy because its sampled h values alternate between 0 and $2/5$, giving $T_{2M}h = 1/5$. Grid symmetry gives $T_{2M}[(1-t)h] = \frac{1}{2}T_{2M}h = 1/10$, which recovers Equation 20. This does not make doubling a universally sufficient rule. For any fixed finite list of uniform interval counts, choosing M as a common multiple places every grid in Equation 19. The example supplies no derivative bound to control the variation that the grids do not resolve.

Exact integration and perturbation quality are different questions. For the displayed $M = 128$, $h(1/2) = 0$, so whole-feature interventions give $\mu(1, 0) = 2/5$ and $\mu(0, 1) = 3/5$. With the same normalized Q response and grid $\{0, 1/2, 1\}$, the aliased feature-2-first ordering has insertion ordinates $(0, 0.84, 1)$ and deletion ordinates $(1, 0.64, 0)$. These give AUCs 0.67 and 0.57. The exact-IG feature-1-first ordering instead has insertion AUC 0.57 and deletion AUC 0.67. The inaccurate map’s ranking therefore scores better on both conventional criteria in this toy: larger insertion AUC and smaller deletion AUC. This occurs because the finite off-path interventions measure a different property from the diagonal path integral. The transverse policy term changes the coordinate gradients without changing these Boolean intervention values. Neither completeness nor exact integration guarantees an optimal perturbation score. This separation is specific to the toy and does not establish RDT ranking efficacy.

The standalone artifact `analysis/revision/nested_grid_aliasing.py` evaluates the unchanged production trapezoidal IG kernel on this CPU toy with $M = 128$, in FP32 and FP64. Budgets 32, 64, 128 reproduce the incorrect order, while 256 reproduces the analytic order to numerical tolerance. The report retains measured coordinate errors, completeness residuals, repeated-run equality and common response curves. It does not replace finite-precision results with the exact equalities above. The two-coordinate example uses no RDT weights or observations. It explains why finite-grid agreement alone cannot certify the integral. It does not explain why any particular PickSingleYCB diagnostic failed or approve a production integration setting.

C Secondary Historical Results

These analyses retain additional observations from the saved campaigns. Their limits on training, reference, context and preprocessing prevent conclusions about model-scale effects, learned-weight-specific efficacy or a causal effect of denoiser contraction.

C.1 The Separately Recorded 1B Cohort

The saved 1B analysis shows why episode weighting matters: changing the weights changes the quantity being estimated. Table 7 uses the same 1,199 calls from 60 episodes for both summaries. Within this cohort, the weighting changes the language summaries more than the vision summaries. The cohort also mixes language-baseline provenance. Neither estimate comes from replicated training or establishes a controlled improvement over 170M.

Table 7: Results from the saved RDT-1B PickCube campaign. Both quantities use 1,199 source calls and 60 episodes. The first gives equal weight to every call. The second gives equal weight to each episode’s median. The conditional 95% episode intervals describe the released mixture. The evaluation seeds share a checkpoint, and the pooled language population includes two recorded baseline provenances.

Estimand	Modality	Insertion AUC	Deletion AUC
Median over calls	Vision	0.940 [0.926, 0.954]	0.288 [0.250, 0.312]
Median over calls	Language	0.541 [0.449, 0.605]	0.267 [0.187, 0.327]
Median of episode medians	Vision	0.940 [0.927, 0.959]	0.289 [0.235, 0.313]
Median of episode medians	Language	0.488 [0.368, 0.581]	0.297 [0.151, 0.342]

C.2 Randomization and Baseline Diagnostics

Table 8 reconstructs the standard verification-campaign correlations. Substantial rank association remains after model randomization, while input shuffling changes the maps more strongly. This contrast does not show that model dependence is unnecessary, and it does not certify semantic grounding. A map based only on input magnitude can also change under input shuffling while ignoring learned policy weights. Appendix E reports the saved frozen-reference and cascade variants, including a sensitivity analysis of dependence grouping.

Table 8: Results from Month 7 model randomization (C1) and input shuffle (C2), using all recorded calls. The table shows median Spearman correlation with conditional 95% episode intervals. The checks measure different forms of dependence. Low C2 correlation does not establish dependence on learned model weights, while C1 changes the integration budget as well as the weights.

Task	Check	Calls	Episodes	Vision ρ	Language ρ
PegInsertion	C1	100	16	0.960 [0.949, 0.969]	0.959 [0.947, 0.966]
PegInsertion	C2	100	16	0.089 [0.067, 0.111]	0.018 [0.004, 0.036]
PickCube	C1	100	26	0.956 [0.948, 0.961]	0.926 [0.917, 0.944]
PickCube	C2	100	26	0.142 [0.128, 0.157]	0.117 [0.099, 0.150]
PickYCB	C1	100	26	0.943 [0.930, 0.949]	0.917 [0.903, 0.931]
PickYCB	C2	100	26	0.164 [0.149, 0.175]	0.128 [0.081, 0.155]
StackCube	C1	100	26	0.958 [0.952, 0.965]	0.965 [0.960, 0.969]
StackCube	C2	100	26	0.140 [0.117, 0.154]	0.266 [0.246, 0.279]

The IG multiplier ($x_i - x'_i$) allows shared structure, but it does not force a particular ranking. Path gradients can dominate the result. At the same time, the input features themselves come from retained learned encoders/adaptors. Persistent correlations therefore motivate input-only and path-gradient-only controls. They do not establish that the multiplier dominates the result. Spearman correlation cannot be interpreted as a fraction of model-independent information. The baseline comparisons in Appendix E show imperfect agreement. Their small number of groups is insufficient to establish equivalence.

C.3 Action Displacement and Solver Resolution

The quadratic score is a normalized squared displacement, so its numerical magnitude depends directly on the chosen convention. Removing $1/D$ multiplies exact scores and attributions by D , while exact ranks, normalized AUC and relative completeness remain unchanged. This algebra does not determine whether an action change matters operationally. That judgment requires action units and a justified behavioral criterion.

The saved solver analysis compares median relative action displacement under top-5% vision deletion between $T = 2$ and $T = 20$. Figure 2 reports the endpoint contrast using conditional resampling of nominal episode pairs. The reference actions vary across solver settings, and exact contexts/noise cannot be authenticated. The intervals allow appreciable increases as well as decreases. Changing T changes the discretization of

the full denoising horizon. It does not add steps along one fixed trajectory. These observations therefore establish neither a causal contraction mechanism nor its absence.

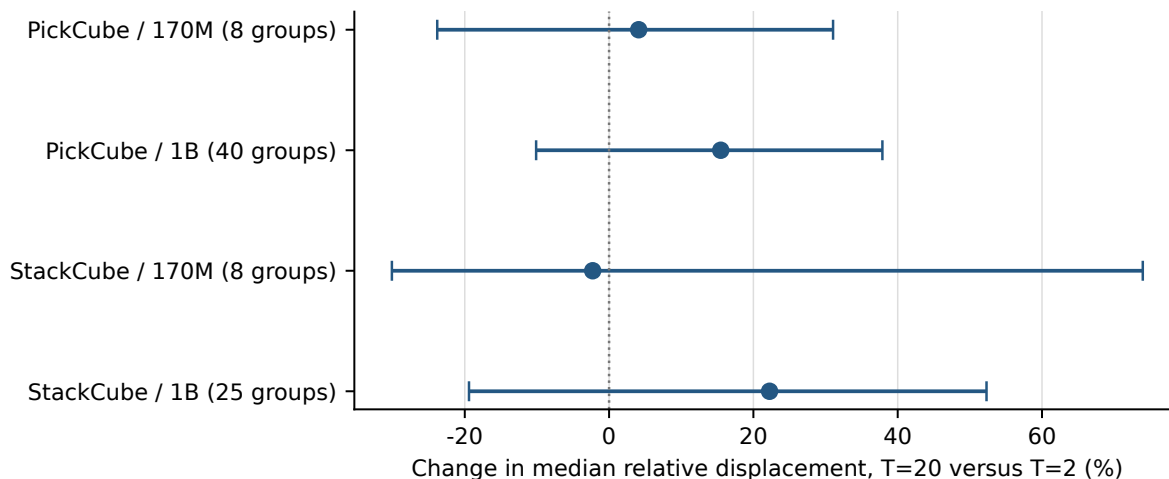


Figure 2: Retrospective change in median relative displacement under top-5% vision deletion. The intervals use nominal episode pairing across solver resolutions and are conditional on the released records and grouping. The vertical reference marks no endpoint change. It is neither a contraction test nor an equivalence margin. Raw context/noise identity remains unknown.

D Exploratory Episode Influence

Large finite response changes affect the paired summaries in Table 2. To locate this sensitivity, we omit each of the 30 recorded episodes in turn, retain all calls from the other 29 episodes, and recompute their equally weighted episode mean. Table 9 reports every case. This analysis followed an exploratory audit of the same records. Its separately saved protocol is therefore not an outcome-blind preregistration. The omission ranges describe deterministic influence. They are not confidence intervals and do not establish that the recorded episodes form an independent sample. The full untrimmed means, medians and conditional intervals remain unchanged.

Table 9: Exploratory omissions of one complete recorded episode. The range covers all 30 omission means. “Same sign” counts agreement with the full-cohort mean. Each case starts with 750 calls in 30 episodes, and rows within a ranking cohort reuse calls. The ranges are not confidence intervals. The separate influence artifact supplies all 240 omission records and their retained-population hashes.

Ranking	Modality	Curve	Full mean	Omission-mean range	Same sign
Q	vision	insertion	-17.0966	[-17.6910, -11.5667]	30/30
Q	vision	deletion	0.1227	[0.1208, 0.1272]	30/30
Q	lang	insertion	-0.1819	[-0.1937, -0.0932]	30/30
Q	lang	deletion	-0.0095	[-0.0158, 0.0182]	27/30
L2	vision	insertion	-18.5620	[-19.2071, -11.9674]	30/30
L2	vision	deletion	0.0184	[0.0128, 0.1264]	30/30
L2	lang	insertion	-0.0329	[-0.0399, -0.0193]	30/30
L2	lang	deletion	-1.3710	[-1.4246, 0.0328]	29/30

All four negative insertion means remain negative after every individual omission. The Q language-deletion mean changes sign in 3 of 30 omissions, and the L2 language-deletion mean changes sign in 1 of 30. Both already have conditional bootstrap intervals spanning zero. In the latter case, the most negative episode

contributes 97.29% of the sum of magnitudes of negative episode means. This measures concentration among negative episode means, not a share of the signed net mean. It does not justify excluding that episode.

Endpoint normalization also helps explain the extreme vision-insertion changes. For the most negative Q-ranking call, the baseline squared residual is 0.0174195, the smallest in its 750-call cohort. At the same time, the maximum saved perturbed residual is 653.053, giving a ratio of about 37,490. The finite Q endpoint gap is 1.7011×10^{-5} , above the historical 10^{-9} cutoff. The native Q AUC is -4464.9082 , while its L2 rescore is -26.0696 . This call combines a small baseline discrepancy with a large intervention response. The division is neither undefined nor by a machine-zero value. These quantities also do not calibrate physical importance in action units. The artifact provides the same diagnostics within each separate ranking cohort. Matching nominal episode labels cannot authenticate a cross-arm pair.

E Additional Retrospective Diagnostics

The frozen-reference and backbone-cascade analyses retain learned encoders/adaptors, and their historical context identities remain uncertain. Table 10 reports recorded-episode grouping and conditional shared-context grouping. Comparing them measures sensitivity to grouping, but does not prove that either grouping is independent. Table 11 reports baseline-ranking point summaries. Their small group count prevents an equivalence conclusion.

Table 10: Results from the saved backbone-randomization variants. The shared-context grouping joins nominal task and episode keys across recorded seeds to check sensitivity to dependence. It does not establish that the inputs were shared. Each interval is conditional on its stated grouping. Learned encoders and adaptors remain in place, while sidecar reuse and language-baseline deviations remain unresolved.

Variant	Grouping	Calls	Groups	Vision ρ	Language ρ
Frozen	Recorded episodes	400	16	0.929 [0.905, 0.940]	0.862 [0.835, 0.879]
Frozen	Shared-context sensitivity	400	8	0.929 [0.904, 0.947]	0.862 [0.820, 0.886]
Cascade	Recorded episodes	400	16	0.713 [0.696, 0.738]	0.910 [0.896, 0.920]
Cascade	Shared-context sensitivity	400	8	0.713 [0.693, 0.763]	0.910 [0.894, 0.926]

Table 11: Correlations between the saved baseline rankings. Only two nominal episode groups are available for each task. The analysis artifact retains interval outputs to describe the resampling results, but we do not present them here as calibrated uncertainty. A positive correlation alone does not mean that the baselines can be used interchangeably.

Task	Calls	Groups	Black/gray	Black/blur	Gray/blur
PickCube	30	2	0.825	0.539	0.455
StackCube	30	2	0.814	0.546	0.496

F Saved Integration-Budget Cohorts

Table 12 replaces the earlier table of unrecovered and duplicated constants with later cohorts that can be reconstructed. This historical sensitivity analysis explicitly retains the norm-selected population and occurrence rule. These calls are not paired with the earlier $m = 64$ main pass on identical observations, baselines, runtime and noise. Truncating trajectories to the same call depth does not create such pairs. An integration-budget effect therefore needs to be evaluated on authenticated fixed contexts, with precision and quadrature controlled.

Table 12: Results for the $m = 128$ cohorts that can be reconstructed. The conditional intervals describe the recorded cohorts. They do not show convergence under paired contexts. The final row removes every conflicting repeated key. The missing original PickCube and StackCube subsets have not been reconstructed or replaced with constants.

Task	Seed	Calls	Groups	Vision residual $\leq 3\%$ (%)
PegInsertion	42	189	8	64.0 [50.3, 76.9]
PegInsertion	142	160	7	76.2 [67.1, 85.0]
PickYCB	42	194	8	73.7 [62.1, 85.5]
PickYCB	142	195	8	71.8 [60.9, 83.1]
PegInsertion (conflicts excluded)	142	142	7	75.4 [63.8, 85.0]